

CS57300
PURDUE UNIVERSITY

DATA MINING

PROPERTIES OF ESTIMATORS

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- ▶ Let $\hat{\theta}$ be an estimate for a population parameter θ
- ▶ Using different samples D will result in different estimates $\hat{\theta}_D$
- ▶ Thus $\hat{\theta}$ is a random variable with a distribution, mean, and variance
 - ▶ We can evaluate the quality of an estimator for θ based on the properties of the sampling distribution of $\hat{\theta}$

BIAS

- ▶ The best estimators produce values that center around the population parameter

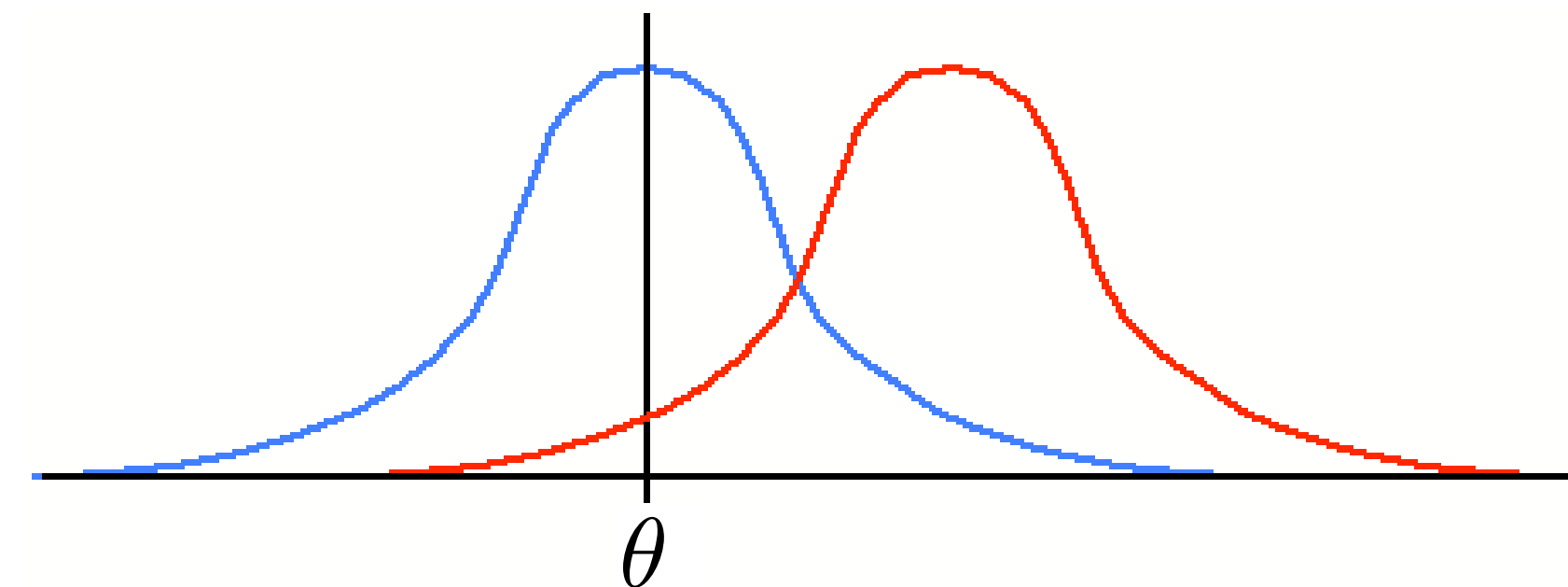
- ▶ The **bias** of an estimator is defined as: $Bias(\hat{\theta}) = E[\hat{\theta}] - \theta$

$E[\hat{\theta}]$
*Average
estimated
parameter*

—

θ
*True
parameter
in popul.*

- ▶ An estimator is unbiased if: $E[\hat{\theta}] - \theta = 0$

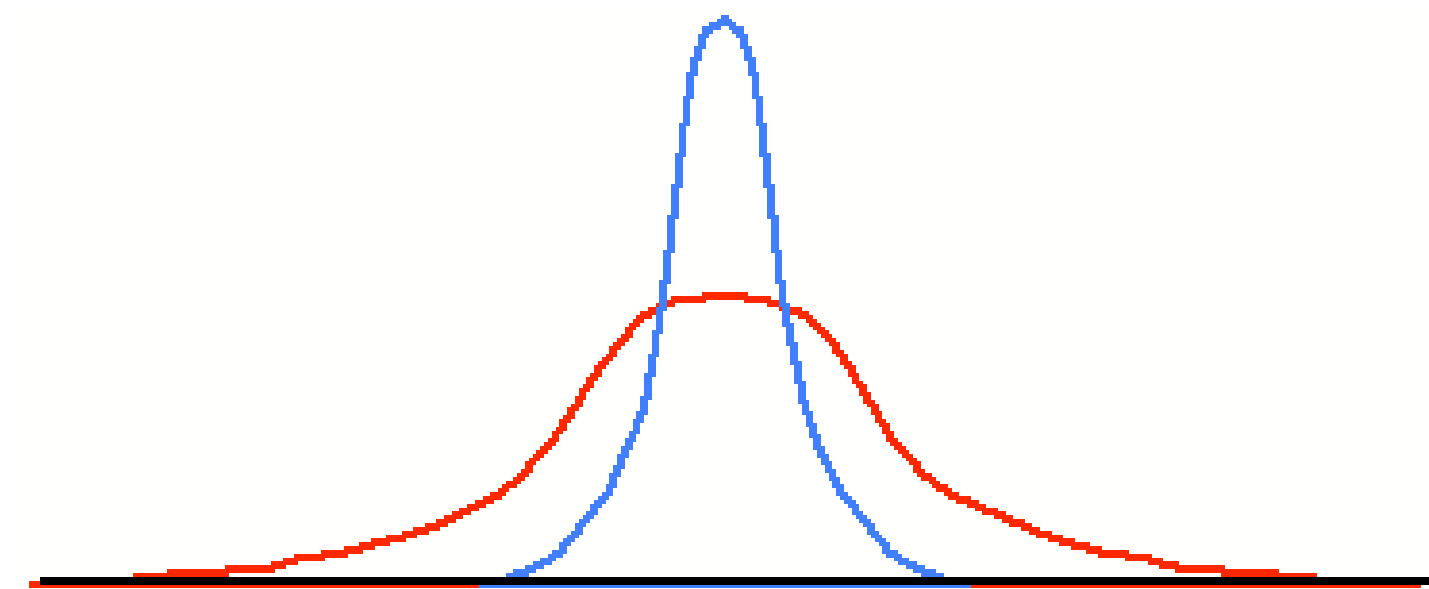


VARIANCE

- ▶ The best estimators produce values that differ only slightly from the population parameter
- ▶ The **variance** of an estimator is defined as: $Var(\hat{\theta}) = E[(\hat{\theta} - E[\hat{\theta}])^2]$

Single
parameter
estimate

Average
estimated
parameter
- ▶ Measures how sensitive the estimator is to different datasets
- ▶ Unbiased estimators with minimum variance are called *best unbiased estimators*



EXAMPLE

- ▶ Ignore data and declare that: $\hat{\theta} = 1.0$
- ▶ Estimate will not depend on data, thus: $Var(\hat{\theta}) = 0$
- ▶ However, in most cases this estimator will have a large bias (non-zero)

BIAS-VARIANCE TRADEOFF

- ▶ The mean-squared error (MSE) of $\hat{\theta}$ is:

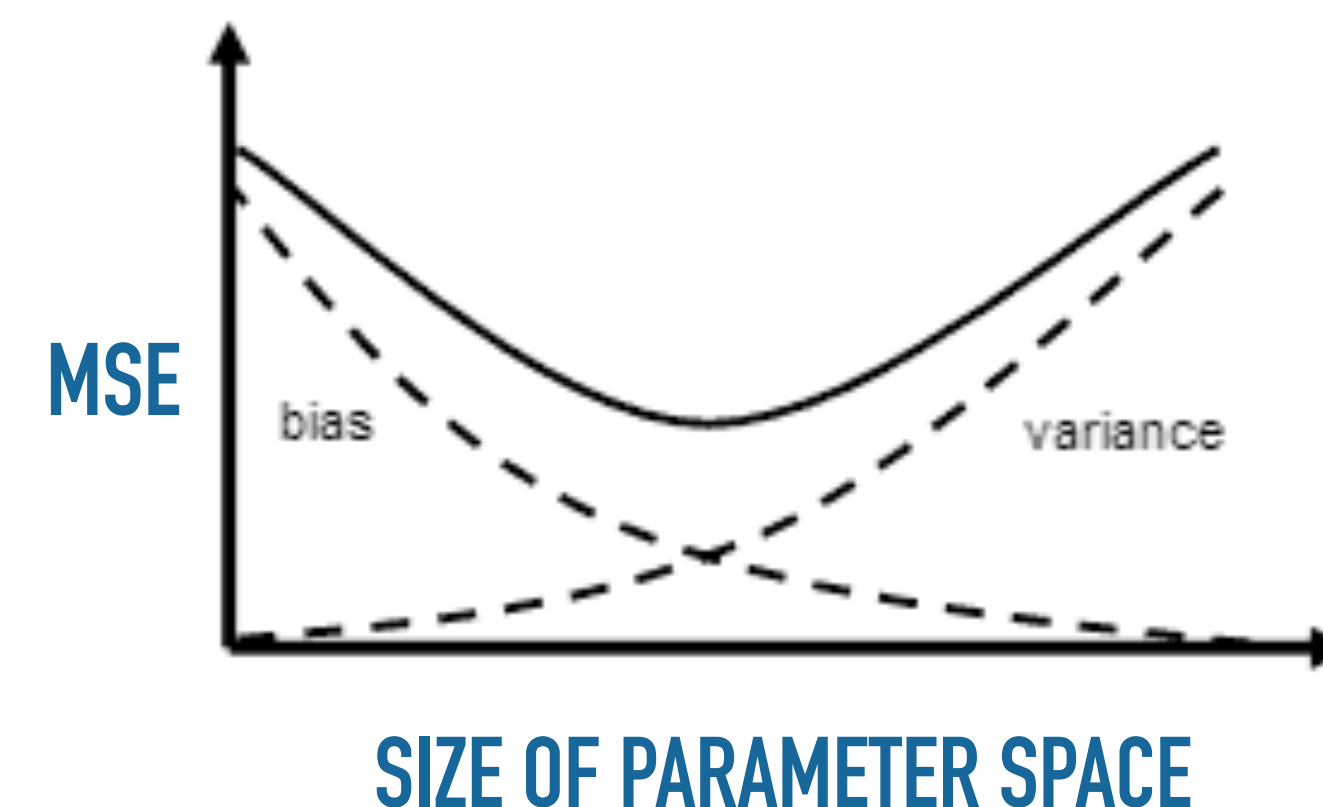
$$E[(\hat{\theta} - \theta)^2]$$

BIAS-VARIANCE TRADEOFF

- ▶ The mean-squared error (MSE) of $\hat{\theta}$ is:

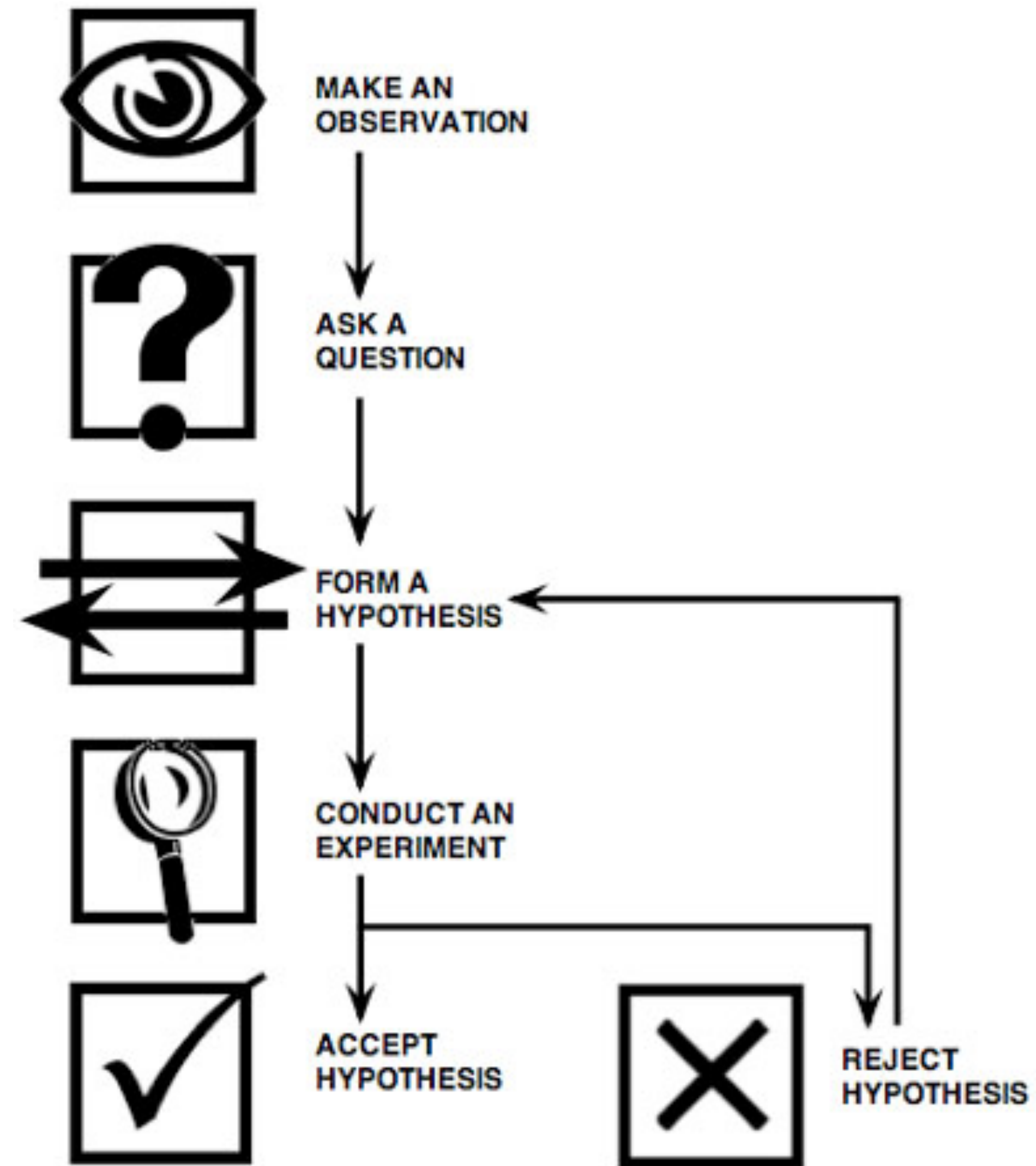
$$\begin{aligned} E[(\hat{\theta} - \theta)^2] &= E[(\hat{\theta} - E[\hat{\theta}] + E[\hat{\theta}] - \theta)^2] \\ &= \underbrace{(E[\hat{\theta}] - \theta)^2}_{\text{bias}} + \underbrace{E[(\hat{\theta} - E[\hat{\theta}])^2]}_{\text{variance}} \end{aligned}$$

- ▶ MSE measures systematic bias and random variance between estimate and population value
- ▶ Tradeoff: reducing bias tends to increase variance and vice versa



HYPOTHESIS TESTING

SCIENTIFIC METHOD



TYPES OF HYPOTHESES

Broad categories

- ▶ **Descriptive**: propositions that describe a characteristic of an object
- ▶ **Relational**: propositions that describe relationship between 2+ variables
- ▶ **Causal**: propositions that describe the effect of one variable on another

Specific characteristics

- ▶ **Non-directional**: an differential outcome is anticipated but the specific nature of it is not known (e.g., the tuning parameter will affect algorithm performance)
- ▶ **Directional**: a specific outcome is anticipated (e.g., the use of pruning will increase accuracy of models compared to no pruning)

**Descriptive
Hypothesis**

**Non-Directional
Relational Hypothesis**

**Directional
Relational Hypothesis**

**Directional
Causal Hypothesis**

Stronger

HYPOTHESES EXAMPLE

- ▶ The query response time is measured for a few different search engines
- ▶ Different hypotheses
 - ▶ **Descriptive:** The query response time for Google follows a normal distribution
 - ▶ **Non-directional relational:** The average response time for a new search engine, QuickSearch, is different from Google's average response time
 - ▶ **Directional relational:** The average response time of QuickSearch is shorter than that of Google's
 - ▶ **Directional causal:** The response time of QuickSearch is shorter than Google's because they cache results of more queries

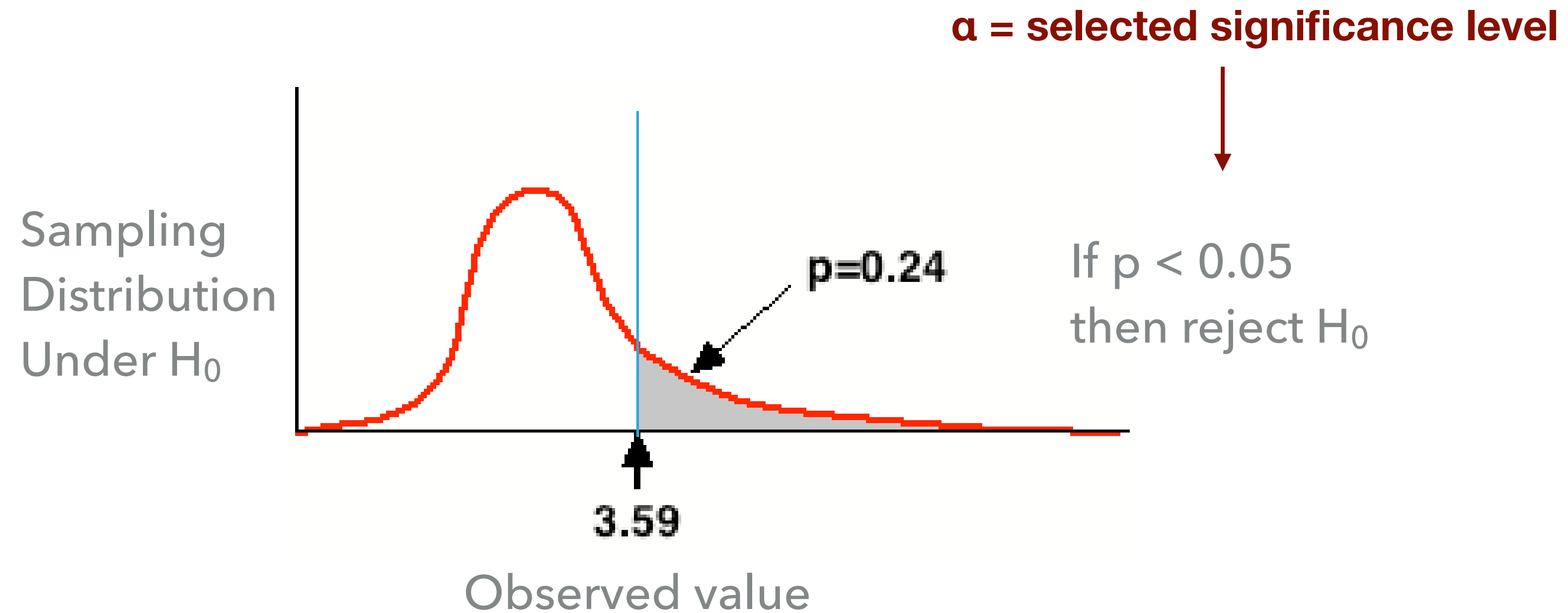
HYPOTHESIS TESTING

- ▶ Statistical hypothesis test is a method used in statistics that tells you the likelihood of a specific result would happen by chance
- ▶ **Null hypothesis** (H_0):
 - ▶ Presumed true until statistical inference indicates otherwise; set up to be refuted by alternative
- ▶ **Alternative hypothesis** (H_1):
 - ▶ Rival hypothesis; that we conjecture is true
- ▶ Assuming the null hypothesis is true, what's the probability of getting a statistic that is at least as extreme as the statistic that was actually obtained through the data?

HYPOTHESIS TESTING STRATEGY

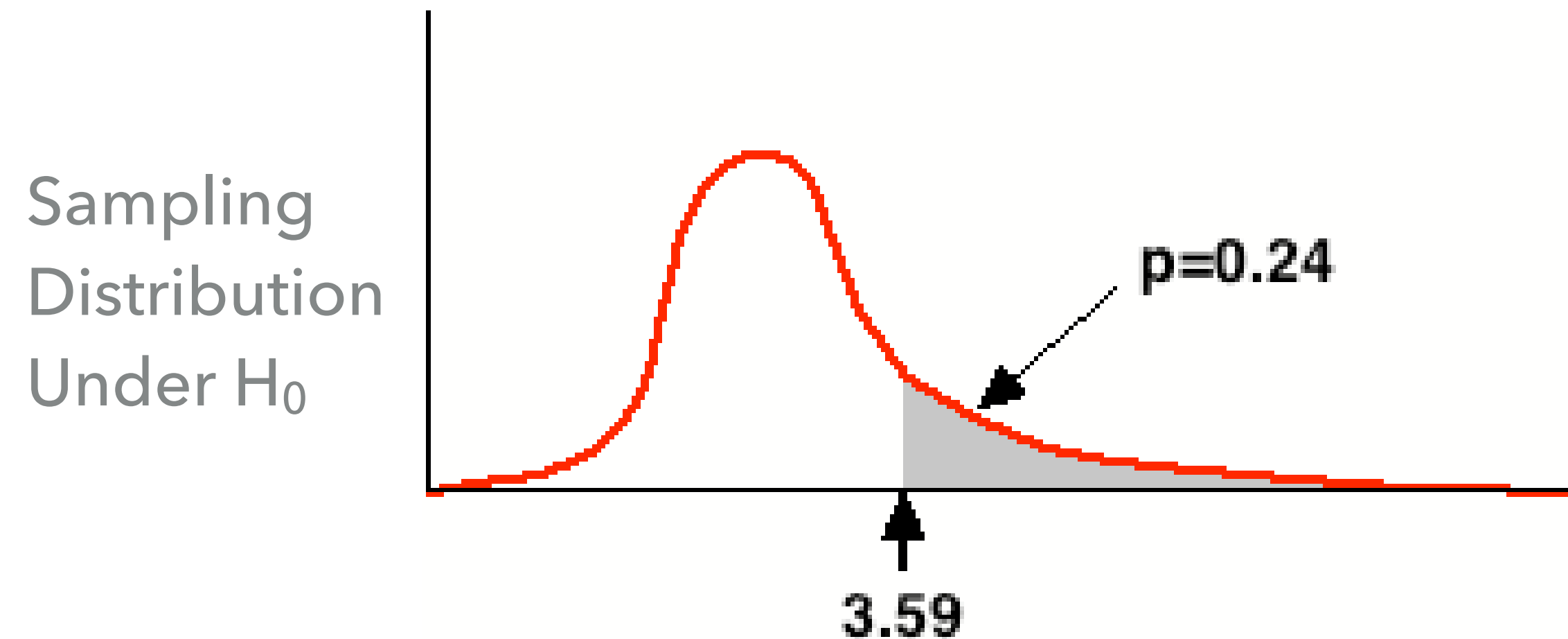
- ▶ Formulate null and alternative hypothesis
 - ▶ H_0 : QuickSearch' mean response time = Google's mean response time
 - ▶ H_1 : QuickSearch' mean response time $\neq$ Google's mean response time
- ▶ Gather a sample statistic (e.g., δ = difference of QuickSearch's and Google's mean response time)
- ▶ Determine the sampling distribution for the statistic under the null hypothesis
- ▶ Use the sampling distribution to calculate the probability of obtaining the observed value of δ , given H_0
 - ▶ If the probability is low, reject H_0 in favor of H_1

REJECTING THE NULL HYPOTHESIS



STATISTICAL SIGNIFICANCE

- ▶ A value of a statistic is **statistically significant** if it is unlikely to occur under the null hypothesis



**significance
level**

$$\alpha = p(\text{reject } H_0 | H_0 \text{ true}) = p(\text{type 1 error})$$

ERRORS

		Decision	
		Reject H_0	Don't reject H_0
Truth	H_0	<i>Type 1 error</i>	
	H_1		<i>Type 2 error</i>

- ▶ Type 1: null is rejected when it is true
 - ▶ E.g., conclude cancer drug increases life expectancy when in fact it doesn't
 - ▶ Generally considered to be most serious error
- ▶ Type 2: null is accepted when it is false
 - ▶ E.g., conclude that cancer drug does not increase life expectancy when in fact it does